

A Preregistered Falsification Program for a Coherence-Control Framework: Four Confirmed Boundary Predictions and Three Falsified Claims

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July 2026 · Companion record and appendices: DOI [10.5281/zenodo.21246247](https://doi.org/10.5281/zenodo.21246247) · Code: github.com/derekhone/uip-phase1-testbeds

Abstract. We report the complete Phase 1 evaluation of a proposed coherence-control framework ("UIP"), conducted as a preregistered falsification program: every claim was stated with numeric pass/fail thresholds before its test was run, no post-hoc re-tuning was permitted after a failure, and all outcomes — confirming and falsifying — are published in full. Across seven numerical testbeds built on standard open-quantum-system dynamics (dephasing and amplitude-damping channels, partial-SWAP coupling, measurement-based partial re-preparation), the framework produced four confirmed quantitative predictions: a steady-state drift fixed point predicted analytically within 2% of simulation; a control-strategy crossover point located in advance; a zero-free-parameter functional law $s^*(\gamma) = \arcsin(\sqrt{\gamma(1-\bar{C}^*)/\bar{C}^*})$ for the crossover coupling strength, confirmed across a γ -sweep with $r = 0.976$; and generalization of the restricted-access advantage to a three-qubit chain. Three claims were falsified at their preregistered thresholds: inheritance-guided correction lost to a static baseline when the target was directly correctable (−11.5%); the framework's decision quantities fell 12.19% (and, with a preregistered predictive extension, 22.06%) short of a beam-search near-optimal adversary, refuting the claim that they are near-sufficient statistics for optimal control; and the constant $\bar{C}^* = 1/1.45$ failed to transfer to a structure-matched classical oscillator system, where the crossover does not exist at all, retiring the framework's universality ("unified") claim. All results are consistent with standard quantum mechanics and feedback control; no new physics is claimed. What survives is a quantum-feedback boundary-predicting bookkeeping method — and, we argue, a demonstration that preregistered kill-conditions are practical and productive for evaluating speculative frameworks by independent researchers.

1. Introduction

Speculative theoretical frameworks proposed outside institutional science typically fail in a characteristic way: not by losing a fair test, but by never taking one. Claims are stated without thresholds, tests are adjusted after results are seen, and failures are silently absorbed into revised versions. This paper documents an attempt to do the opposite with one such framework.

UIP ("inheritance physics") was proposed as a general account of coherence, its transfer between systems ("inheritance"), its loss ("drift" and "waste"), and its recovery by adaptive correction, organized around a schematic balance equation $dC/dt = G(R,U) - L(D,W)$. At the outset, the framework carried broad cross-domain ambitions. Phase 1 posed a narrow question: *when the framework's objects are given exact numerical definitions in a standard open-quantum-system testbed, does it predict anything — and can it be killed?*

The evaluation protocol was fixed in advance and held throughout:

- Every claim was preregistered with explicit numeric pass/fail thresholds before its test was run.
- Falsification conditions ("kill-conditions") were stated alongside each prediction.
- After a failure, at most one preregistered revision of the failed hypothesis was permitted; iteration until success was prohibited as curve-fitting.
- All results, confirming and falsifying, are published (Appendices A–G of the companion record).

2. Common testbed and definitions

All quantum experiments use standard tools: single- and few-qubit density matrices; dephasing channels (phase-flip Kraus form, rate γ) and amplitude damping; partial-SWAP coupling $U = \exp(-is(I - \text{SWAP}))$ between qubits, with per-step transfer fraction $\tau = \sin^2(s)$; and measurement-based partial re-preparation toward a target state ($\rho \rightarrow (1-f)\rho + f\rho_{\text{target}} \otimes \rho_{\text{rest}}$) as the correction primitive.

The framework's objects were given exact definitions:

- **Coherence:** $C(\rho) = \ell_1$ -norm coherence (sum of magnitudes of off-diagonal elements; $2|\rho_{01}|$ for a qubit).
- **Drift:** $D(t)$ = trace distance from the target state.
- **Resonance:** R = normalized phase alignment of off-diagonal elements between coupled subsystems.
- **Gain/Loss:** G and L instantiated as explicit benefit/cost expressions per testbed, specified before running.
- **Sensing noise:** where policies act on estimated quantities, Gaussian noise was added to estimates ($\sigma = 0.02$ on coherence, $\sigma = 0.01$ – 0.02 on rates), with 30 seeds per condition.

3. Confirmed predictions

3.1 Steady-state fixed point (Appendix A)

Single qubit holding $|+\rangle$ under dephasing and amplitude damping, with measurement-based adaptive correction. A key negative-then-positive result: unitary echo correction alone could not satisfy the framework's recovery tests; only re-preparation ("recirculation") could, because coherence lost to the environment cannot be restored unitarily. With recirculation, the steady-state drift fixed point was predicted analytically as $D^* = 0.1032$ and simulated at 0.1051 — agreement within 2%. Contraction toward the fixed point ($k < 1$) held in the transient regime at moderate gain.

3.2 Restricted-access boundary condition (Appendix C)

Following the falsification reported in §4.1, a sharpened boundary hypothesis was preregistered: inheritance-guided correction is advantageous *only when the target is not directly correctable*. In a two-qubit system where only upstream qubit A is accessible and target B is shielded, the inheritance-guided policy (v3) beat all baselines as predicted, confirming the boundary condition rather than the original unrestricted claim.

3.3 Crossover point and zero-free-parameter law (Appendices D–E)

The framework's benefit/cost bookkeeping implies a crossover coupling strength s^* below which correction of the accessible relay costs more transferable coherence than it delivers. Setting delivered benefit equal to disturbance cost yields a steady relay coherence threshold $\bar{C}^*$ and

$$s^*(\gamma) = \arcsin(\sqrt{(\gamma(1-\bar{C}^*)/\bar{C}^*)}), \quad \bar{C}^* = 1/1.45.$$

The crossover location was first predicted and confirmed at a single operating point (Appendix D), then the full functional form — with no free parameters — was confirmed across a sweep of dephasing rates with correlation $r = 0.976$ between predicted and measured crossover points, each within the preregistered ± 0.05 window (Appendix E).

3.4 Generalization (Appendix E)

The restricted-access advantage and boundary structure carried over to a three-qubit chain (A accessible $\rightarrow$ relay $\rightarrow$ shielded target), satisfying the preregistered generalization criterion.

4. Falsified claims

4.1 Inheritance-guided control with an accessible target (Appendix B)

Claim: the inheritance quantity $I(A \rightarrow B)$ identifies where a limited correction budget should be spent.

Result: in a two-qubit system with both qubits accessible, the inheritance-guided policy (v1) underperformed the static always-correct-the-target baseline by 11.5% in mean target coherence, with worse worst-case performance; a preregistered revision (v2, $I_2 = R \cdot C(A) \cdot \tau_{\text{eff}}$) passed its weak-coupling consistency prediction but failed its advantage prediction at every tested coupling strength.

Verdict: falsified. Diagnosis: the rule over-valued upstream coherence relative to the transfer-rate bottleneck; when the target is directly correctable, protecting it directly dominates. This failure produced the boundary hypothesis confirmed in §3.2.

4.2 Sufficiency of the framework's decision quantities (Appendix F)

Claim: the quantities $\{C(A), R, \hat{\gamma}\}$ are near-sufficient statistics for optimal coherence-preserving control (preregistered: within 1% of optimal; kill-condition: gap $> 5\%$). **Adversary:** a near-optimal policy found by beam search (width 200) with single-flip hill-climbing, holding full knowledge of the dynamics; since the dynamics are deterministic, this lower-bounds true optimal performance, and any tightening by reviewers widens the reported gap. **Result:** optimal mean $C(B) = 0.5257$ vs v3 = 0.4616 — a 12.19% gap. The optimal policy exploits anticipatory burst structure phase-locked to the noise schedule; the framework's instantaneous quantities carry no timing information. A single preregistered extension (v4: an online drift-schedule estimator inferred from decay history) was tested and performed worse (0.4097; 22.06% gap): knowing when the noise strikes is not enough to reproduce genuine lookahead optimization. **Verdict: falsified twice; no further iterations run.** Closing the gap requires standard dynamic programming, with no framework content.

4.3 Cross-platform universality of $\bar{C}^*$ (Appendix G)

Claim: the constant $\bar{C}^* = 1/1.45$ and the law $s^*(\gamma)$ transfer to a structure-matched classical system (two noisy oscillators, beat-exchange coupling with $\tau = \sin^2(s)$, phase-diffusion decay $e^{-\gamma}$, partial signal re-injection as correction). **Preregistered predictions:** classical crossovers within ± 0.05 of $s^*(0.15) = 0.2628$ and $s^*(0.25) = 0.3420$. **Result:** no crossover exists at any tested coupling; always-correct dominates everywhere. The quantum crossover arises from measurement back-action — correcting the relay disturbs the off-diagonal structure being transferred — and classical re-injection has no such cost. The constant is not merely different off-platform; the quantity it balances is undefined there. **Verdict: falsified; the universality claim, and with it the framework's "unified" designation, is retired.** Cross-domain applications (biological, cosmological, energetic) proposed in early versions of the framework are unsupported by this record and positively contraindicated by this result.

5. Summary of the Phase 1 ledger

Claim	Preregistered threshold	Outcome
Steady-state fixed point D^* (analytic vs simulated)	quantitative agreement	CONFIRMED (<2%)
Inheritance-guided control, accessible target (v_1 , v_2)	beat all baselines	FALSIFIED (~11.5%)
Restricted-access boundary condition (v_3)	beat all baselines under shielding	CONFIRMED
Crossover point s^* located in advance	± 0.05 window	CONFIRMED
Zero-free-parameter law $s^*(\gamma)$; 3-qubit generalization	± 0.05 per point	CONFIRMED ($r = 0.976$)
Decision-quantity sufficiency vs optimal adversary (v_3 , v_4)	gap < 1%; kill at > 5%	FALSIFIED (12.19%, 22.06%)
Cross-platform transfer of $\bar{C}^* = 1/1.45$	± 0.05 per point	FALSIFIED (no crossover exists)

6. Discussion

6.1 What survives

Within quantum feedback control, the framework's bookkeeping located fixed points, crossovers, and regime boundaries in advance — including one functional law with zero free parameters. These are real, reproducible, preregistered quantitative predictions. They are also fully consistent with standard quantum mechanics and feedback control theory: on a proof ladder from 0 (pure speculation) to 7 (new physical principle), the record places the framework at Level 4 — confirmed predictions inside known theory — with the paths to Levels 5 and 7 tested and closed at this evidence class.

6.2 What died, and why that is the point

The framework's three most ambitious claims — that its inheritance quantity guides action, that its quantities suffice for near-optimal control, and that its constant is universal — were each tested at a preregistered threshold and each failed. In every case the failure was mechanistically diagnosable and produced a sharper true statement than the claim it replaced. The final framework is smaller than the initial one, and truer.

6.3 Methodological claim

Independent of the framework's fate, this record demonstrates that a full preregistered falsification program — numeric kill-conditions, bounded revision, published negative results — is practical for a single independent researcher using open tools, and that it is *productive*: two of the four confirmed results (§3.2, §3.3) were discovered only because a falsification forced a boundary to be drawn. We suggest this protocol as a template for evaluating speculative frameworks generally, where the common alternative is unbounded post-hoc revision.

6.4 Limitations

All results are numerical simulations built by the framework's author; no independent replication or hardware validation exists yet. The optimal-policy adversary is a lower bound (beam search, not exact DP). The classical analog in §4.3, while structure-matched, is one of many possible classical instantiations — though rescuing transfer by adding a hand-built disturbance cost would import the quantum feature under test and would not constitute independent transfer.

7. Open invitations

- **Break the law:** test $s^*(\gamma)$ outside the swept range, under non-Markovian or amplitude-damping-dominant noise. A preregistration for hardware validation on transmon qubits (measured T2, predicted crossover, $\pm$ windows) accompanies the repository.
- **Tighten the adversary:** exact dynamic programming or a POMDP solution under sensing noise; any improvement widens the reported Appendix F gaps and strengthens the negative result.
- **Replicate or refute:** all code (numpy/scipy, seeded, deterministic) is public. Refutations are as welcome as replications and will be acknowledged in any revision.

Data and code availability

Complete appendices A–G, all simulation code, and preregistration texts: DOI [10.5281/zenodo.21246247](https://doi.org/10.5281/zenodo.21246247) and github.com/derekhone/uip-phase1-testbeds.

Acknowledgments

The author acknowledges AI assistance (Abacus.AI ChatLLM) in simulation construction, adversarial review, and manuscript preparation; all claims, thresholds, and the decision to publish every failure are the author's responsibility. This work was conducted under a single standing rule, older than the method it produced: "*A false balance is an abomination to the LORD, but a just weight is His delight*" (Proverbs 11:1). Soli Deo gloria.